ICT Express 12 (2026) 324–329

Contents lists available at ScienceDirect

ICT Express

journal homepage: www.elsevier.com/locate/icte

Optimized hybrid CNN-LSTM-GRU model for cyber attack detection using Adaptive Equilibrium Optimization

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ARTICLE INFO

Keywords:
Cyber attack detection
Deep learning
CNN-LSTM-GRU
Adaptive Equilibrium Optimization
Network security

ABSTRACT

Modern network security has a great difficulty in cyber attack detection. In this context, this work presents an CNN-LSTM-GRU Model for cyber attack detection. AEO was used to maximize feature selection for the model, therefore lowering unnecessary data while maintaining important attack patterns. With 98% accuracy, the suggested model beats conventional GRU, LSTM, and RNN architectures according to experimental data. Computationally efficient for the proposed model achieves equivalent accuracy to the Transformer model with less number of FLOPs and parameters.

1. Introduction

Particularly considering the growing complexity of cyber threats, cyber attack detection has become a crucial topic of study. Deep learning (DL) models have become quite effective tools for improving the detection and categorization of cyberattacks in different fields, including IoT, wireless sensor networks, and industrial control systems. The approaches, success, and difficulties of applying deep learning for cyberattack detection are investigated in this synthesis.

Particularly those using architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), deep learning models have shown great potential in identifying cyber threats. CNNs have been very useful in identifying Distributed Denial of Service (DDoS) assaults, for example, when models were trained on real-time traffic data to highly accurately classify attack types [1]. In time-series data, RNNs have also been used to examine sequential data, thereby fitting for spotting trends suggestive of cyber dangers [2]. A major benefit of deep learning over conventional machine learning approaches is its capacity to automatically extract features from unprocessed data without much human feature engineering.

The paper proposes an optimized hybrid CNN-LSTM-GRU model for cyber attack detection, integrating Adaptive Equilibrium Optimization (AEO) for feature selection and SMOTE for class balancing, achieving high accuracy while significantly reducing computational complexity compared to Transformer-based models.

2. Related work

Recent studies have proposed various methods for the detection of cyber attacks, particularly DDoS attacks. Zhou et al. [3] introduced a statistical approach to protect healthcare services against DDoS during the COVID-19 pandemic, while Mishra et al. [4] combined statistical

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https://doi.org/10.1016/j.icte.2025.09.006
Received 16 April 2025; Received in revised form 13 June 2025; Accepted 20 September 2025
Available online 14 October 2025
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analysis with machine learning to predict DDoS variants. Setia et al. [5] applied machine learning to detect DDoS attacks in VANET cloud environments, addressing the dynamic nature of vehicular networks.

Beyond DDoS detection, Sharma et al. [6] proposed a federated learning-based model to improve privacy in critical infrastructure systems. Mehla et al. [7] and Upadhyay et al. [8] explored secure data sharing in cloud banking and access control in metaverse environments, respectively. Furthermore, Vijayakumar et al. [9] demonstrated the role of deep reinforcement learning in improving pedestrian and vehicle safety.

Although these works highlight the growing use of AI in cybersecurity, they often rely on single-model architectures or are domain specific. In contrast, our proposed Optimized Hybrid CNN-LSTM-GRU model integrates CNNs, LSTMs, and GRUs with Adaptive Equilibrium Optimization (AEO) to deliver a more robust and generalizable solution for cyber attack detection across diverse environments.

3. Proposed work

In this section, details of the proposed approach are provided. As shown in Fig. 1, the architecture consists of three primary blocks: Backbone Block, Neck Block, and Head Layer. Data preparation starts the Backbone Block; thereafter, AEO-based feature selection removes duplicated characteristics while maintaining important attack patterns. Using a concurrent CNN-LSTM-GRU architecture, The Neck Block effectively analyzes sequential patterns while CNN extracts spatial data and LSTM gathers long-term dependencies. Following an adaptive pooling mechanism in the Head Layer, the retrieved features are refined; ultimate prediction comes from a fully connected classifier. The CNN block uses a 1D convolutional layer with 64 filters (kernel size 3, padding 1), batch normalization, and ReLU activation to extract local spatial features. The LSTM and GRU blocks each consist of 2 layers with 64 hidden units to model sequential and long-term dependencies.

3.1. Data normalization

Since the dataset contains features with different numerical scales, Min-Max Normalization is applied to bring all values within the range $[0, 1]$:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

where:

- X is the raw feature value,
- X_{min} and X_{max} are the minimum and maximum values of the feature,
- X_{norm} is the normalized feature value.

3.2. Data balancing using SMOTE

SMOTE generates synthetic samples for minority classes using k-nearest neighbors. For a given minority class sample x_i, a new synthetic sample x_{new} is generated as:

$$x_{new} = x_i + \lambda \cdot (x_{nn} - x_i) \tag{2}$$

where:

- x_i is an original minority sample,
- x_{nn} is one of its k-nearest neighbors,
- λ is a random number in $[0, 1]$, ensuring variability.

3.3. Feature selection using adaptive equilibrium optimization (AEO)

Feature selection is performed using AEO, which updates the position of search agents in the optimization space as follows:

$$C_i^{new} = C_{eq} + (C_i - C_{eq}) \cdot F_i + GCF_i \cdot V \cdot (1 - F_i) \tag{3}$$

where:

- C_{eq} is the equilibrium candidate chosen from the equilibrium pool,
- C_i is the current feature vector,
- F_i is the control factor for exploration and exploitation,
- GCF_i is the generation control factor,
- V is the velocity component.

The fitness function for optimal feature selection is defined as:

$$\text{Fitness} = \arg\min_t \sum_{i=1}^{K+1} S_i(M_i) \tag{4}$$

where $S_i(M_i)$ represents the entropy measure for different feature subsets.

3.4. Feature extraction using CNN-LSTM-GRU

3.4.1. CNN feature extraction

Feature extraction using a convolutional layer:

$$F^{(l)} = \sigma\left(W^{(l)} * X + b^{(l)}\right) \tag{5}$$

where:

- $W^{(l)}$ is the convolutional filter,
- X is the input feature map,
- $b^{(l)}$ is the bias,
- σ is the activation function (ReLU).

Max pooling operation:

$$P^{(l)} = \max\left(F^{(l)}\right) \tag{6}$$

3.4.2. LSTM feature extraction

The Long Short-Term Memory (LSTM) network processes sequential input data by updating its internal states using a memory cell. The updated cell state at time step t is formulated as:

$$C_t = \gamma_f \circ C_{t-1} + \gamma_i \circ \tilde{C}_t \tag{7}$$

where:

$$\gamma_f = \psi(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget Gate)} \tag{8}$$

$$\gamma_i = \psi(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input Gate)} \tag{9}$$

$$\tilde{C}_t = \phi(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(Candidate Cell State)} \tag{10}$$

$$\gamma_o = \psi(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(Output Gate)} \tag{11}$$

$$h_t = \gamma_o \circ \phi(C_t) \tag{12}$$

Here, $\psi(\cdot)$ represents the sigmoid activation function, $\phi(\cdot)$ represents the hyperbolic tangent activation function, and $\circ$ denotes element-wise multiplication.

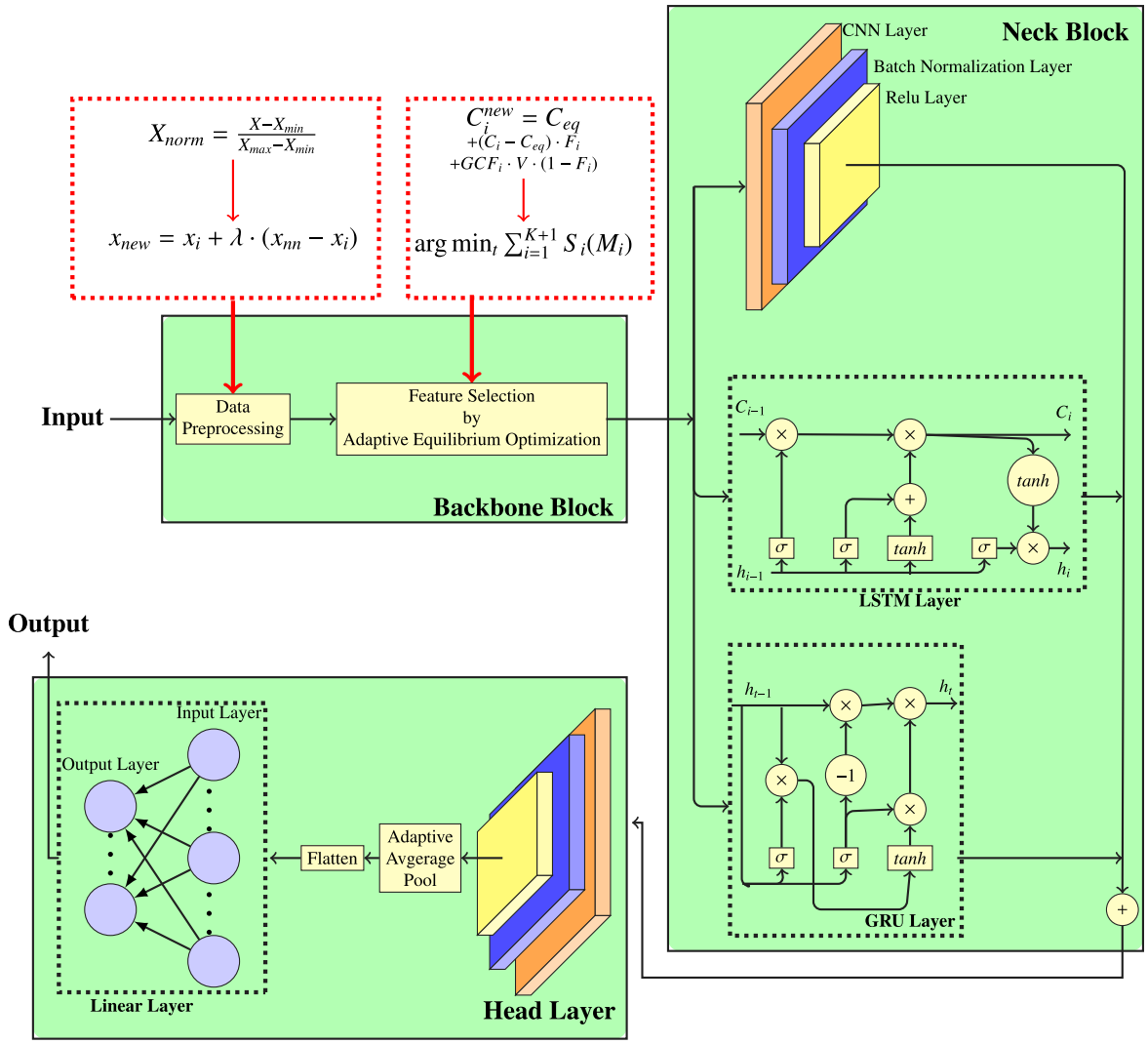


Fig. 1. Proposed Approach.

3.4.3. GRU feature extraction

The Gated Recurrent Unit (GRU) employs an update mechanism that selectively retains information across time steps. The update process is defined as:

$$\zeta_t = \psi(W_z \cdot [h_{t-1}, x_t] + b_z) \quad \text{(Update Gate)} \tag{13}$$

$$\gamma_t = \psi(W_r \cdot [h_{t-1}, x_t] + b_r) \quad \text{(Reset Gate)} \tag{14}$$

$$\tilde{h}_t = \phi(W_h \cdot [\gamma_t \circ h_{t-1}, x_t] + b_h) \tag{15}$$

$$h_t = (1 - \zeta_t) \circ h_{t-1} + \zeta_t \circ \tilde{h}_t \tag{16}$$

where: - ζ_t controls how much past information is retained, - γ_t determines the influence of past hidden states, - $\tilde{h}_t$ is the candidate activation, - $\psi(\cdot)$ and $\phi(\cdot)$ are activation functions ensuring non-linearity.

The final extracted feature set is obtained by concatenating CNN, LSTM, and GRU outputs:

$$F_{final} = P^{(l)} \oplus h_t^{LSTM} \oplus h_t^{GRU} \tag{17}$$

where $\oplus$ denotes concatenation.

3.5. Classification using fully connected layer

Adaptive average pooling is applied:

$$A = \frac{1}{n} \sum_{i=1}^{n} F_i \tag{18}$$

The fully connected layer outputs:

$$y = \text{softmax}(W_f \cdot A + b_f) \tag{19}$$

where:

- W_f and b_f are trainable parameters,
- y is the probability distribution over classes.

The classification is optimized using the categorical cross-entropy loss:

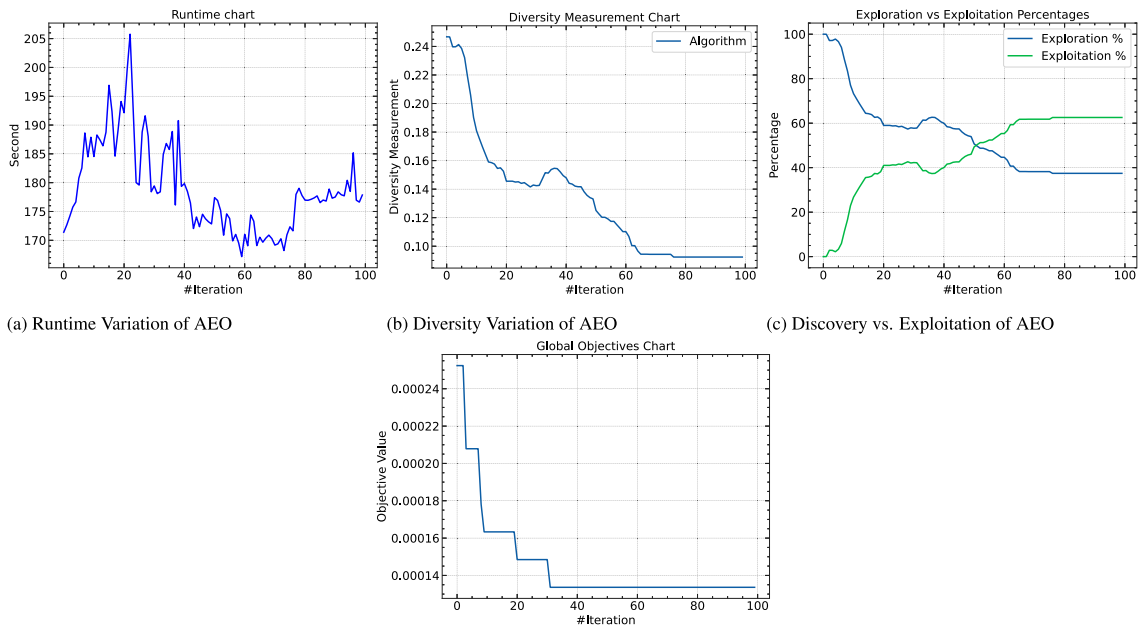
$$\mathcal{L} = -\sum_i y_i \log(\hat{y}_i) \tag{20}$$

where:

- y_i is the ground truth label,
- $\hat{y}_i$ is the predicted probability.

The final optimization step:

$$\theta^* = \arg\min_\theta \mathcal{L}(\theta) \tag{21}$$



(a) Runtime Variation of AEO (b) Diversity Variation of AEO (c) Discovery vs. Exploitation of AEO

(d) Global Fitness Variation of AEO

Fig. 2. Performance of AEO.

4. Results and discussion

4.1. Dataset preprocessing

The performance of the proposed model was evaluated using the well-known NSL-KDD data set for intrusion detection research. This dataset has five different classes: Benign, DoS Attacks, Probe Attacks, U2R, and Sybil. However, a notable class imbalance is seen, whereby some attacks form contains far fewer samples than others. Original distribution consisted of 67,351 Benign samples, 45,927 DoS Attack samples, 11,656 Probe Attack samples, 995 U2R samples, and just 43 Sybil samples. This imbalance can result in biassed classification performance since models might prefer the majority classes while trying to precisely identify the minority classes. The Synthetic Minority Oversampling Technique (SMOTE) was used to provide synthetic samples for underrepresented classes in training dataset.

4.2. Performance of AEO

The AEO algorithm is used to identify the important features in the dataset. Initially, the data set has 41 results, and AEO selects 20 important features. Multiple criteria were used to evaluate the efficacy of the AEO algorithm.

Fig. 2(b) shows the running performance of AEO. The runtime shows swings at first, peaked about iteration 20, then declines gradually and stabilizes. This pattern shows that before stabilizing as the search converges toward ideal answers, the algorithm needs a greater computational effort during the early exploratory phase. In Fig. 2(d), the diversity measurement is presented. Starting at the outset of the optimization process, the diversity value guarantees a thorough investigation of all conceivable alternatives. However, as repetitions go on, the variety declines, signifying a change from discovery to exploitation. Although early convergence is prevented, this behavior is necessary to improve the chosen features.

Fig. 2(c) shows the progressive change from exploration to exploitation in the balance between them. The exploration percentage is much larger at the beginning to guarantee a thorough search over the feature space. Gradually, exploitation takes the front stage as the number of iterations rises, enabling the method to converge toward an optimal subset of features by means of solution refinement. The equilibrium reached at iteration 60 points to AEO's effective control of the trade-off between these two important phases. Showed in Fig. 2(a), the convergence of the objective value demonstrates a constant decrease in the optimization goal function over iterations. While the latter rounds show a steady convergence toward an ideal feature set, the dramatic drop in the first iterations emphasizes a fast improvement in the solution quality. This behavior minimizes the objective function, confirming the efficacy of AEO in feature selection.

	Precision	Recall	F1 Score
Bening	0.98	0.97	0.98
DoS Attacks	0.99	1	1
Probe Attacs	0.98	0.99	0.99
U2R	0.97	1	0.98
Sybil	0.99	0.96	0.98
Accuracy	0	0	0.98
Macro Avg	0.98	0.98	0.98
	0.98	0.98	0.98

Fig. 3. Classification Report.

4.3. Model performance evaluation

Three main statistical metrics, classification report, confusion matrix and receiver operating characteristic (ROC), curve, were used to assess the trained model. The classification report is shown in Fig. 3. With precision, recall and F1 score values close to 0.98 the findings show that the model achieved high classification performance across all classes. While the precision numbers validate the model's capacity to minimize false positives, the high recall values show its capacity to properly detect attack forms. Showed in Fig. 4, the confusion matrix offers a comprehensive analysis of examples accurately and incorrectly labeled for each class. Showed in Fig. 5, the ROC curve compares the true positive rate of each class with the false positive rate.

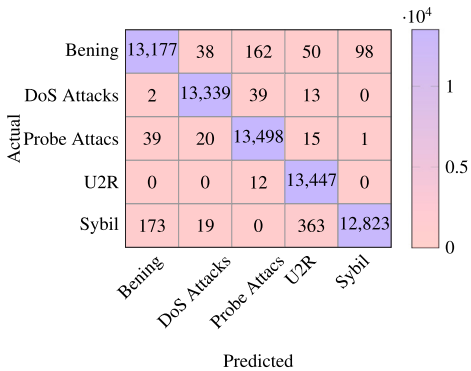


Fig. 4. Confusion Matrix.

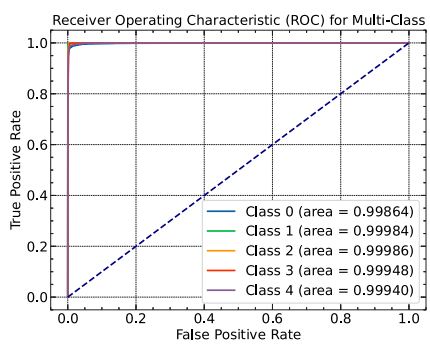


Fig. 5. ROC Curve.

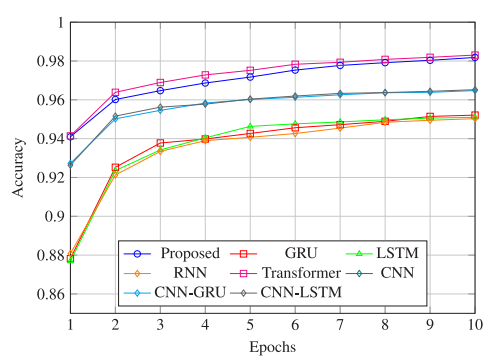


Fig. 6. Accuracy Comparison.

4.4. Comparative analysis

The performance of the proposed model in cyber threat detection was evaluated against different deep learning architectures such as GRU, LSTM, RNN, Transformer, CNN, CNN-GRU and CNN-LSTM. Figs. 6 and Fig. 7 present the precision and loss comparison of the proposed model. Only the transformer model has better accuracy and loss as compared to the proposed model. However, in Fig. 8 it is evident that the proposed model required fewer FLOPS. The suggested model achieves equivalent precision with just 7.24×10^5 FLOPs and 1.2×10^5 parameters, the Transformer model shows a rather higher computational cost requiring 1.79×10^7 FLOPs and 5.65×10^5parameters. T This makes the proposed model more effective than the transformer model.

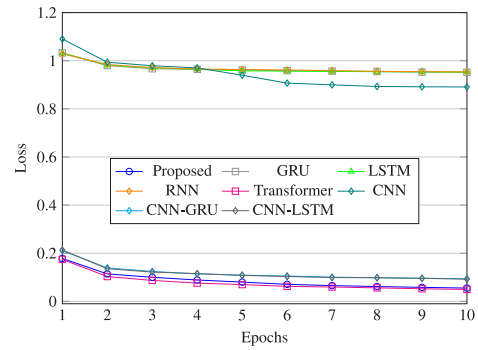


Fig. 7. Loss Comparison.

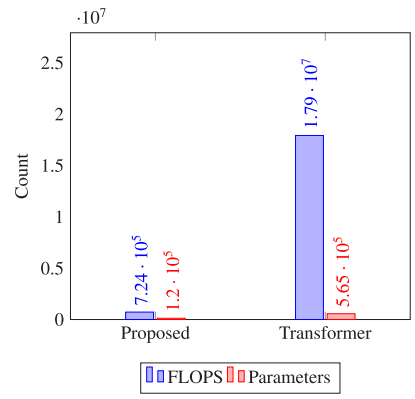


Fig. 8. Comparison With Transformer.

5. Conclusion

This work proposes an optimized hybrid CNN-LSTM-GRU model for cyber attack detection. The proposed model achieves 98% accuracy, surpassing CNN, CNN-GRU, CNN-LSTM, GRU, LSTM and RNN while maintaining competitive performance against the Transformer model. Furthermore, the suggested model has lower FLOPs and parameter count compared to the transformer model. The results develop a computationally strong and efficient method for detecting cyber attacks in the network environment. As part of our future work, we plan to validate the proposed model on more recent and diverse datasets, such as CICIDS2017, CICDDoS2019, and UNSW-NB15. This will help to further assess the robustness and generalization capabilities of the model in different network environments and contemporary attack scenarios.

CRediT authorship contribution statement

Brij B. Gupta: Writing – review & editing, Supervision. **Shin-Hung Pan:** Writing – original draft, Investigation. **Akshat Gaurav:** Validation, Methodology. **Varsha Arya:** Formal analysis, Conceptualization. **Wadee Alhalabi:** Software, Formal analysis, Data curation. **Kwok Tai Chui:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This project was funded by Deanship of Scientific Research (DSR) at King Abdulaziz University, Jeddah under grant No. (IFPDP-261-22), the authors, therefore, acknowledge with thanks DSR technical and financial support.

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